INCH-POUND

MIL-DTL-22442/1C(AS)
5 January 1998
SUPERSEDING
MIL-C-22442/1B(AS)
2 September 1988

DETAIL SPECIFICATION SHEET

CABLE ASSEMBLIES, AIRCRAFT AUDIO, SWITCH ASSEMBLY

This specification is approved for use by the Naval Air Systems Command, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-22442 listed in the issue of the Department of Defense Index of Specifications and Standards (DoDDIS) specified in the solicitation..

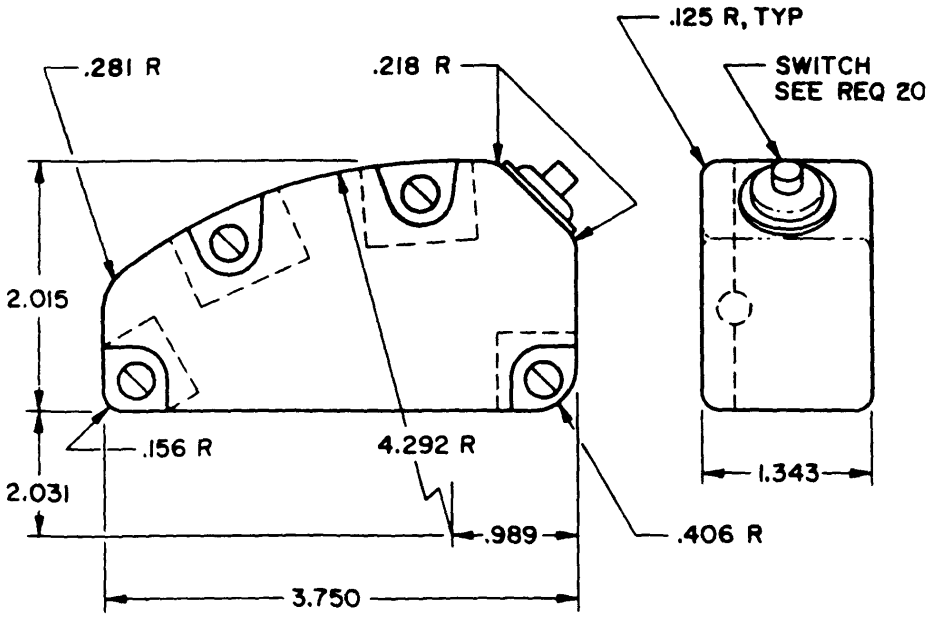
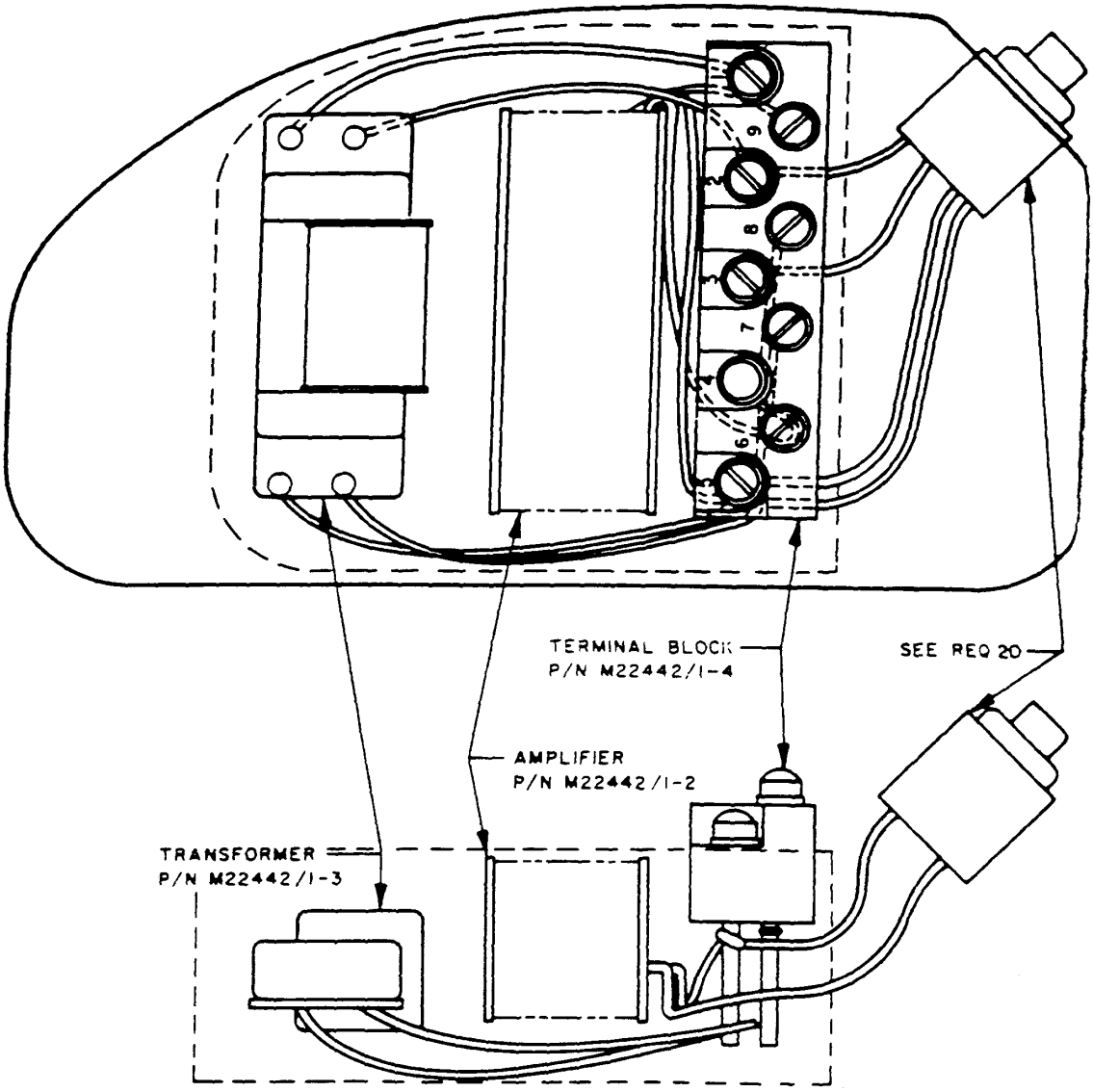


FIGURE l. P/N M22442/l-l.

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NOTE: The components enclosed by dotted lines may be potted.

FIGURE 2. Switch assembly (typical)

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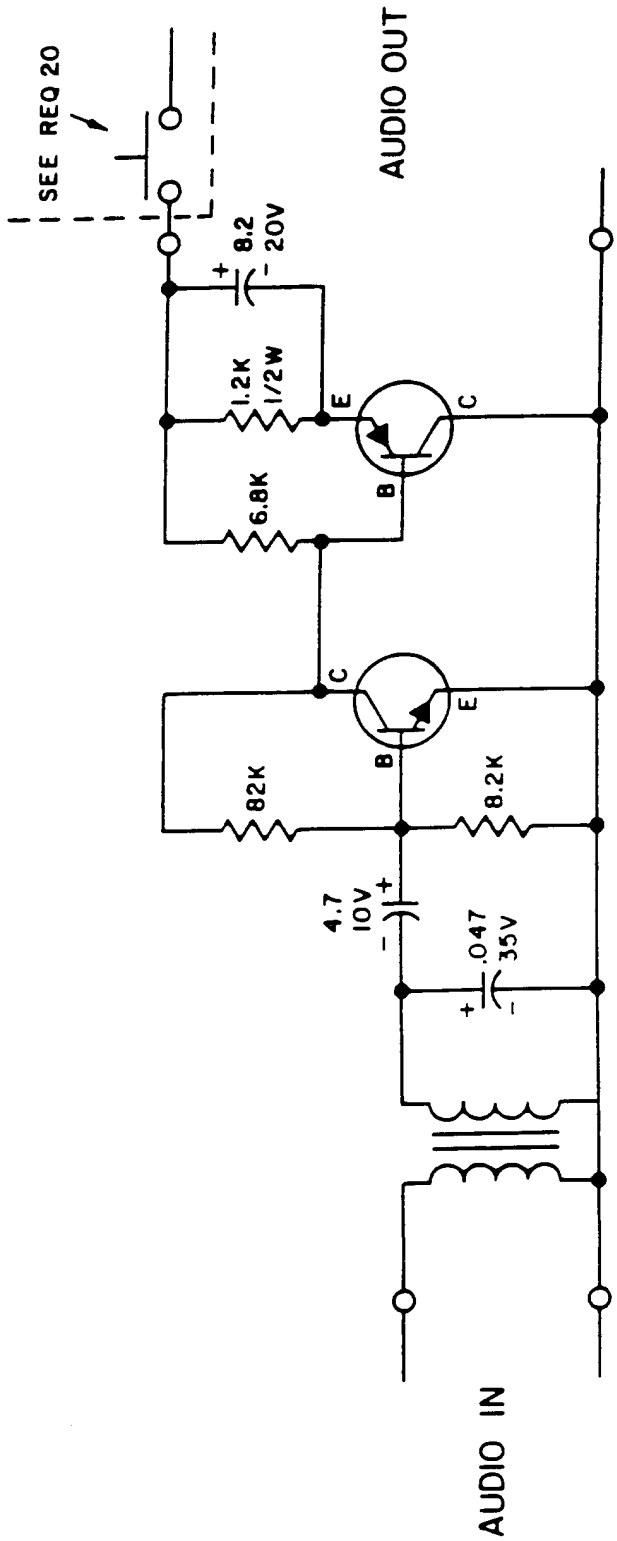


FIGURE 3. Schematic amplifier (typical) P/N M22442/1-2.

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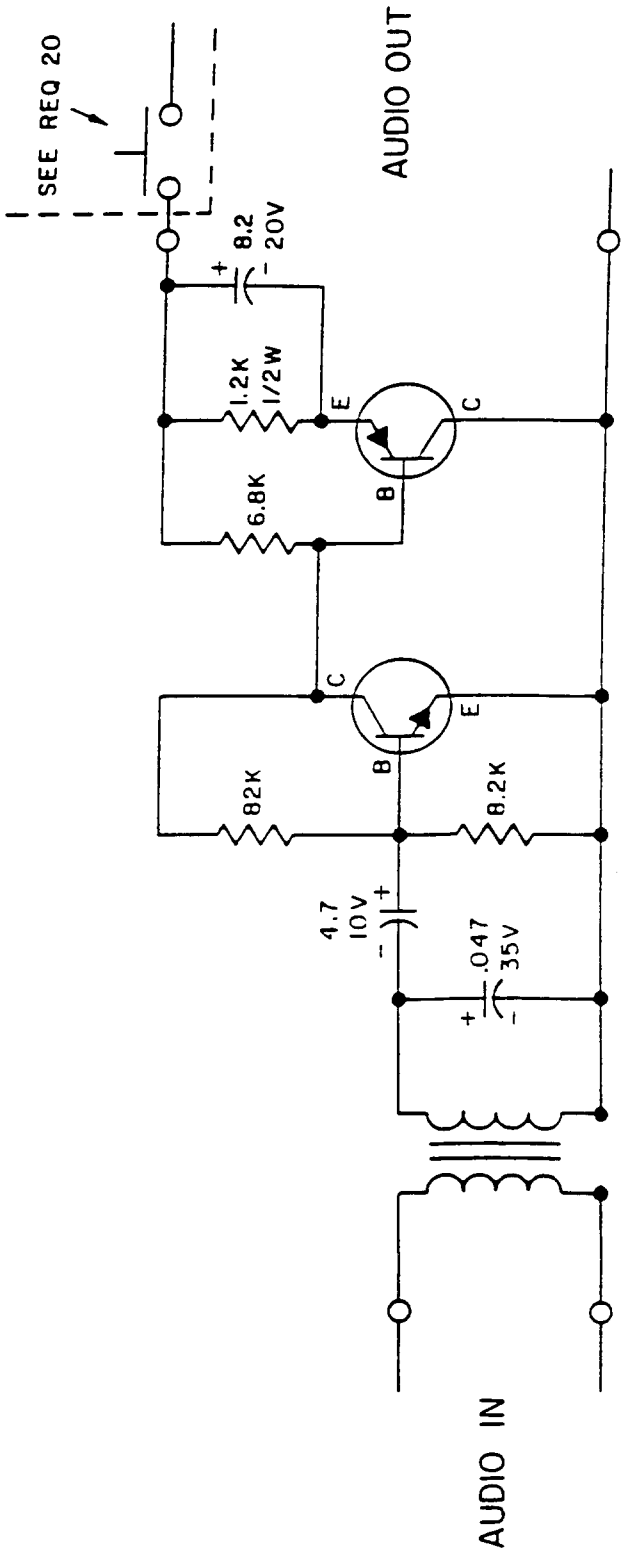


Figure 4 Schematic amplifier (typical) P/N M22442/1-5

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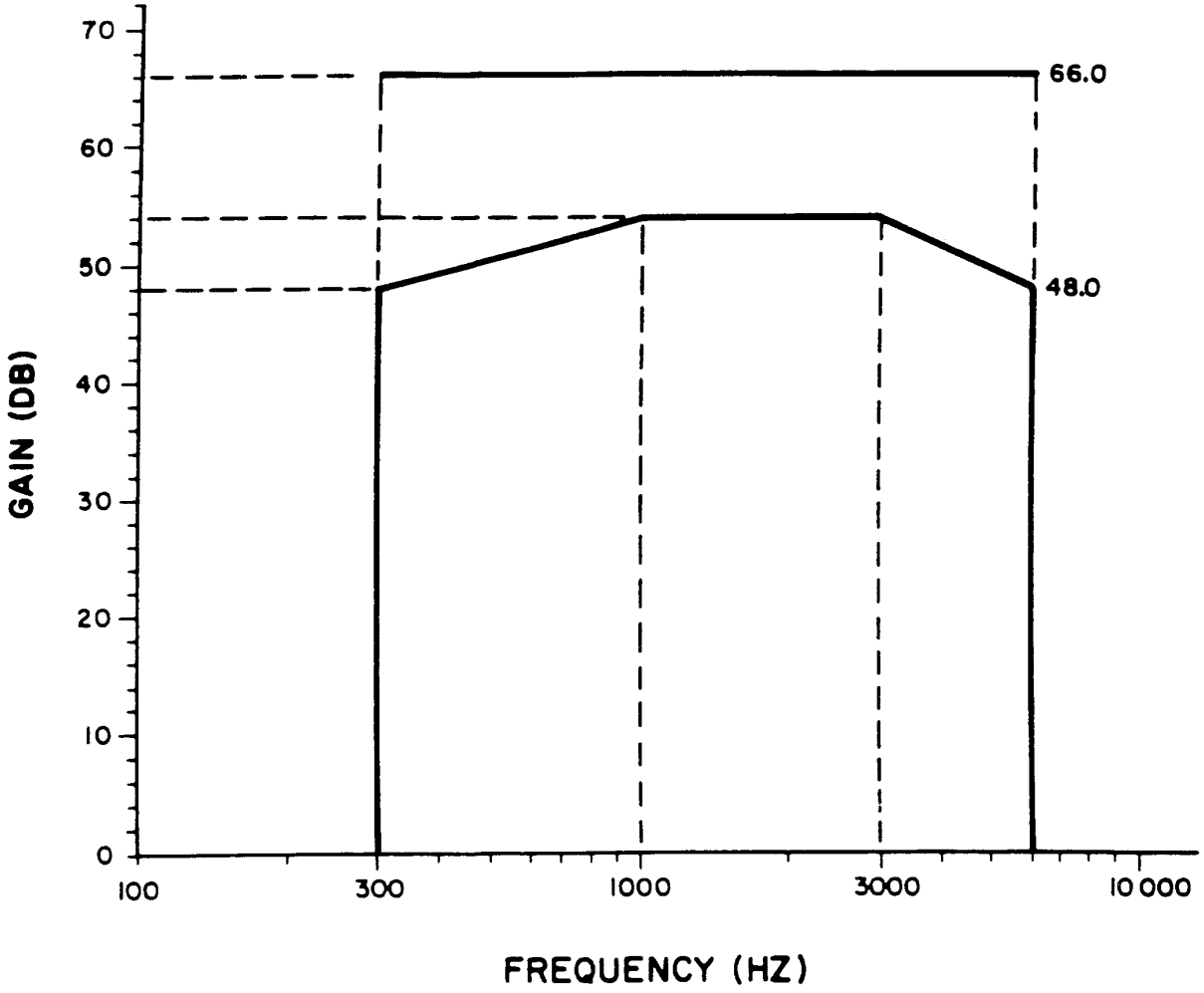


FIGURE 5. M22442/1-2, M22442/1-2-5 Amplifier response limited

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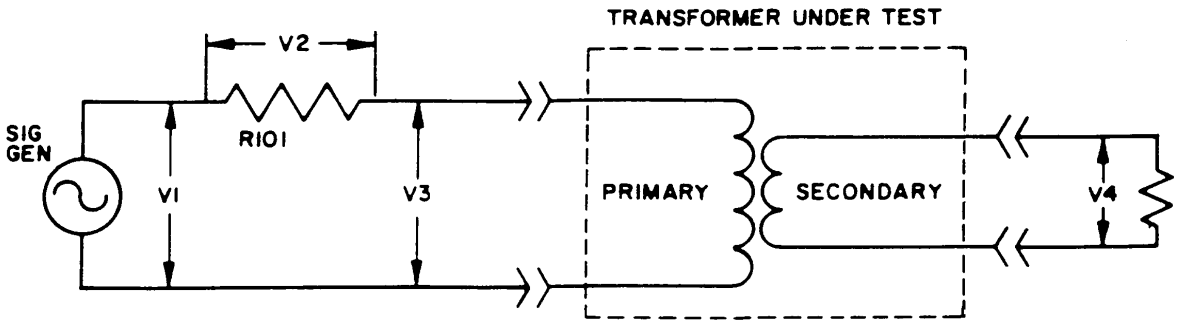
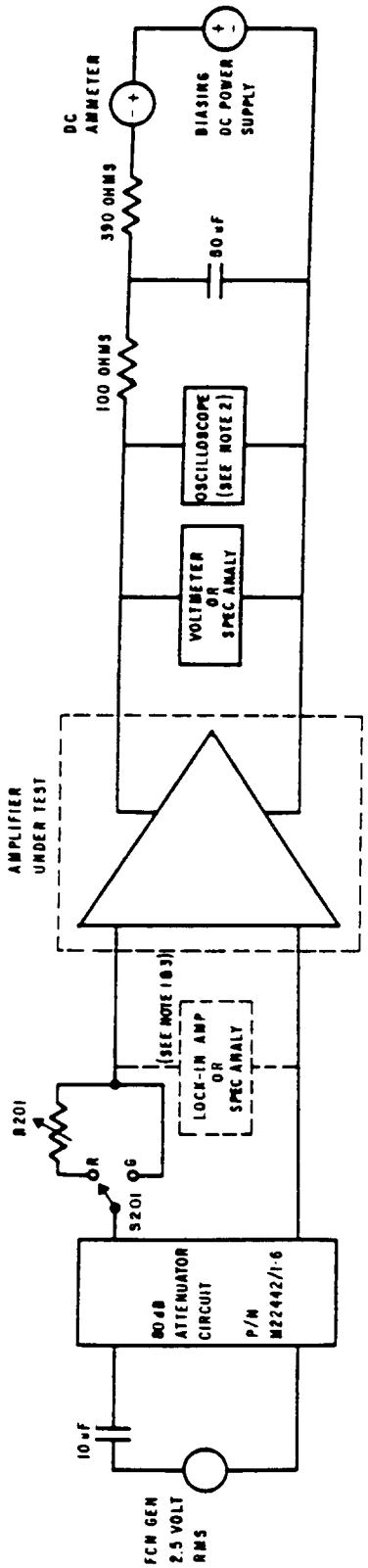


FIGURE 6. Transformer test circuit (see 4.6.3)

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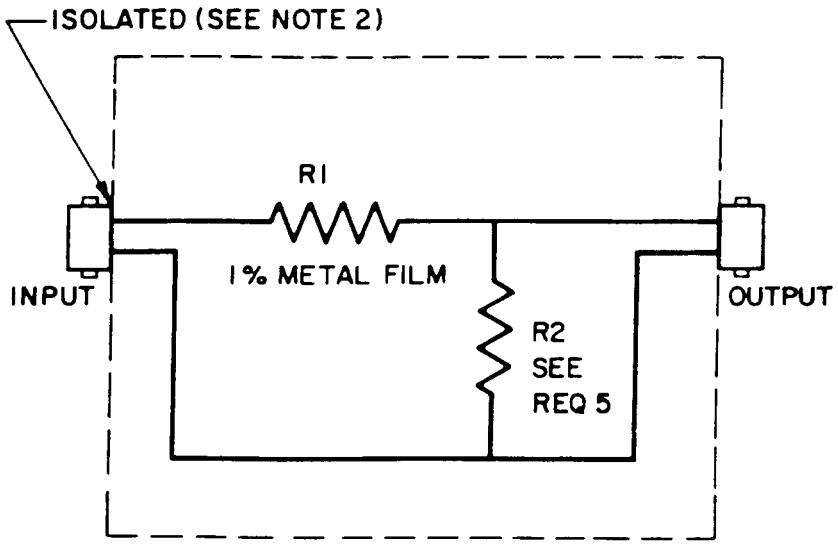


NOTES:

1. Neither the lock-in amplifier nor the spectrum analyzer is required if the attenuator circuit is calibrated.
2. Oscilloscope is required to ascertain if any waveshape distortion, oscillation, etc. is present in the amplifier under test or is being inducted by the test setup.
3. R201 = Decade resistanced box that should meet 6.4.1.5 of the general specification
4. Switch S201 shall be placed in the "R" position when performing the amplifier input impedance test and placed in the "G" position when performing all other tests.

FIGURE 7. Amplifier test circuit.

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NOTES:

l. Case shall be Pomona Electronics, Inc. Model 3320 or equivalent (as approved by the qualifying activity).

2. The existing BNC input connector shall be replaced with isolated BNC type (Amphenol P/N 31-10 or equivalent (as approved by the qualifying activity)).

3. Resistor R1 nominally equals [(A-l)/A] * Zs, where A is a 10,000 to l attenuation, and Zs is the function generator output impedance.

4. Resistor R2 nominally equals (Zs * Zl)/((A * Zl) - Zs), where A is a 10,000 to l attenuation, Zs is the function generator output impedance, and Zl is the nominal load impedance.

5. Resistor R2 shall be calibrated to provide 10,000 to l attenuation. Suggest using a nominal one inch length of manganin resistance wire, of suitable AWG size. For 50 ohm function generator output, suggest using .08926 ohm/foot Driver-Harris AWG #15, .05707" dia, or equivalent (as approved by the procuring activity). For 600 ohm function generator output, suggest using .5678 ohm/foot Driver-Harris AWG #23, .02257" dia, or equivalent (as approved by the procuring activity).

6. R1 shall be of a value such that (R1 + R2)/R2 is 10,000.

7. M22442/l-6 attenuator is described in literature as a fixed asymmetrical L pad attenuator.

FIGURE 8. Attenuator schematic P/N M22442/l-6.

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APPLICABLE DOCUMENTS:

Non-Government Publications. The following documents forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitations (see 6.3).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D4673	-	Standard Classification System for Acrylonitrile-Butadiene-Styrene (ABS) Plastics and Alloys Molding and Extrusion Materials (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa 19103-1187.)

REQUIREMENTS:

l. The switch assembly may contain a microphone amplifier, microphone keying switch, earphone matching transformer and terminal block, or any combination of, as specified in the applicable specification sheet.

* 2. The switch housing and terminal block shall be molded from molding plastic conforming to ASTM-D4673, ABS 121, black or equivalent (as approved by the procuring activity).

3. Impedance matching transformers shall have the following characteristics:

M22442/l-3 impedance matching transformer

a. Input matching impedance: 600 ohms ± 15% at 1000 Hz with a 7.5 ohm resistive load and rated input voltage.

b. DC winding resistance: Primary less than 100 ohms.
Secondary less the 10 ohms.

c. Power rating: 13.4 volts AC RMS applied to primary under load conditions results in 1.3 ± .2 volts AC RMS output.

M22442/l-7 impedance matching transformer

a. Input matching impedance: 600 ohms ± 15% at 1000 Hz with a 15.0 ohm resistive and rating input voltage.

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b. DC winding resistance: Primary less than 100 ohms.
Secondary less than 10 ohms.

c. Power rating: 13.4 volts AC RMS applied to primary under load conditions.

M22442/l-8 impedance matching transformer

a. The secondary winding shall be 16 ohms with a center tap.

b. Input matching impedance: 4000 ohms ± 15% at 1000 Hz with each side of secondary terminated in an 8.0 ohm resistive load and rated input voltage.

c. DC winding resistance: Primary less than 150 ohms.
Secondary less than 10 ohms.

d. Power rating: 6.0 volts AC RMS applied to primary under load conditions.

4. Impedance matching transformers. Impedance matching transformers furnished as part of a cable assembly under this specification shall have primary and secondary windings which are electrically isolated from each other and shall, as a minimum, meet the following requirements under ambient and during environmental conditions. Except during measurement of the dc winding resistance, the secondary winding of the transformer shall be terminated with a non-inductive, high precision (± 1%) resistive load equivalent to the nominal load impedance, and the rated input voltage specified in the specification sheet shall be applied to the primary winding of the transformer.

5. Transformer efficiency. The efficiency of the transformer shall be defined as the ratio of the output power to the input power (P_{out}/P_{in})*l00%. The efficiency of the impedance matching transformer shall be at least 56 percent at 1000 Hz, which is equivalent to a maximum power loss of 2.5 dB.

6. Transformer input matching impedance. The nominal input matching impedance of the transformer shall be as specified.

7. Transformer distortion. The total harmonic distortion (THD) of the transformer shall be less than 15% between 300 Hz to 1000 Hz and less than 5.0 % between 1000 to 3000 Hz.

8. Transformer dc winding resistance. The maximum dc resistance shall be as specified.

9. Transformer output. With the appropriate input signal at l KHz, the output signal shall meet the power ratings as specified for the transformer.

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10. Transformer tests. The following transformer measurements shall be made using the test circuit of figure 6. Rl02 is a non-inductive, high precision (±1%) resistive load equivalent to the nominal load impedance. R_{101} is a non-inductive, high precision resistance whose value shall be chosen to be 40 to 60 percent of the nominal input matching impedance of the transformer. Unless otherwise specified, the measurements shall be performed at 1000 Hz, and the rated input voltage shall be applied across the primary winding of the transformer under test.

a. Transformer efficiency (power) measurement. The input power to the transformer shall be determined using the test circuit of Figure 6 substituting voltage measurements as indicated into the following equation:

$$P_{in} = \frac{(V_1)^2 - (V_2)^2 - (V_3)^2}{2XR_{101}}$$

Then determine the output power by measuring the load voltage V_4 using the equation:

$$P_{out} = \frac{(V_4)^2}{R_{102}}$$

The efficiency of the transformer, $\frac{P_{out}}{P_{in}}$ *100% shall meet the minimum specified in Requirement 5.

b. Transformer input matching impedance measurement. The nominal input matching impedance of the transformer shall be determined at 1000 Hz by substituting the values obtained for V2 and V3 above into the equation.

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Z (input matching impedance) = $\frac{V_3 R_{101}}{V_2}$

c. Transformer distortion measurement. The total harmonic distortion (THD) of the transformer under load conditions shall be determined using a distortion analyzer (see 6.8.1.4), of the general specification or by using a spectrum analyzer (see 6.8.1.3) of the general specification to determine the amplitudes of the harmonics and using the following equation

THD (Total harmonic distortion) = $\sqrt{\sum (A_i / A_f)^2}$ x 100%

Where A_i is the amplitude of the i_{th} harmonic and A_f is the amplitude of the fundamental. All harmonics within the range of 300 to 6000 Hz shall be considered. The distortion shall be measured at fundamental frequencies of 300, 1000, 3000 Hz or as otherwise specified by the qualifying activity to insure compliance with the requirements of Requirement 7.

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d. Transformer dc resistance. The primary and secondary dc resistance's shall be as specified.

e. Transformer output. The transformer output shall be measured to ensure compliance with Requirement 9 by measuring the primary and secondary voltages when terminated with the proper resistive load.

11. Audio amplifier shall have the following characteristics:

Amplifiers M22442/l-2 and M22442/l-5

a. The frequency response of the amplifiers shall be within the limits of figure 5 from 300 to 6000 Hz.

b. The nominal load impedance shall be 100 ohms.

12. Audio amplifiers. Audio amplifiers furnished as part of a cable assembly under this specification shall, as a minimum, meet the following requirements under ambient and during environmental conditions. The output of the amplifier shall be terminated with a non-inductive, high precision (±1%) resistive load equivalent to the nominal load impedance (l00 ohms unless specified), and the nominal input voltage shall be 250 (±10) microvolts.

13. Amplifier frequency response (voltage gain). The gain of the amplifier under load shall be as specified over the frequency range of 300 to 6000 Hz, except that when operated between -35^{0}C and -54^{0}C, the output may be down an additional 6 dB from the gain under standard conditions.

* 14. Amplifier input power. The input power to the amplifier shall be 27.5 (±.5) volts dc and shall be provided across the output terminals of the amplifier. The amplifier shall require no more than 18 milliamps dc input to provide the specified gain over the range of 300 to 6000 Hz.

15. Amplifier distortion. Unless otherwise specified, the total harmonic distortion (THD) of the amplifier shall not exceed 15% between 300 to 1000 Hz and 5.0 % over the frequency range of 1000 to 3000 Hz with any supply voltage from 22 to 30 volts dc. Under low voltage conditions, 10 to 22 volts dc, the amplifier distortion shall not exceed 40% between 1000 to 3000 Hz.

16. Amplifier linearity. The output voltage shall be a linear function of the input voltage within ±2 dB over the input voltage range of 50 to 500 microvolts.

17. Amplifier stability. No damage shall result from amplifier operation with the output terminated in an open circuit, and the amplifier shall resume normal operation once the nominal load impedance is restored. The amplifier shall be stable and shall show no evidence of regeneration or oscillation when operated from a power source which may provide any voltage from 10 to 30 volts, with input and output circuits terminated in the required impedance. There shall be no output when the microphone keying switch is released.

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18. Amplifier impedance. Unless otherwise specified, the input impedance to the amplifier shall be 5 (+10, -2) ohms at 1000 Hz with the output under load.

* 19. Amplifier tests. The following amplifier measurements shall be performed using the test circuit of Figure 7 except that a true RMS ac voltmeter (see 6.4.1.1 of the general specification) which is calibrated to the 50 microvolts level may be substituted for the lock-in amplifier. Unless otherwise specified, the DC bias shall be set to 27.5 ± .5 volts as measured across the DC bias supply terminals when placed in the circuit. Unless otherwise specified, the nominal input voltage to the amplifier shall be 250 (±10) microvolts, and the amplifier output shall be terminated with a non-inductive, high precision (±1%) resistive load equivalent to the nominal load impedance. The input voltage to the amplifier is provided by the M22442/l-6 shielded attenuator circuit, which provides 10,000 to l attenuation (80 dB). The input voltage to the amplifier under test shall be determined by measuring the input voltage to the calibrated M22442/l-6 attenuator and dividing by 10,000. Thus, a function generator output level of 2.5 volts will provide the nominal 250 microvolts to the amplifier under test. Any variation on the test circuit of figure 7 must be approved by the Qualifying Activity (Naval Air Warfare Center, Aircraft Division, Indianapolis) prior to acceptance of the test results.

a. Amplifier set-up procedure. Using the test circuit of figure 7 before the amplifier under test is connected, set switch S20l so that R20l is out of the circuit (S20l shall remain in this position throughout the test sequence until the impedance test procedure of Requirement l9(f) is performed). Adjust the signal generator for a 2.5 volt signal into the shielded attenuator at 1000 Hz. Connect the amplifier under test and perform measurements specified below.

b. Amplifier frequency response (voltage gain) and linearity. The gain of the amplifier shall be determined using the test circuit of figure 7. The output voltage shall be measured using a spectrum analyzer (see 6.8.1.3) of the general specification or voltmeter (see 6.4.1.1) of the general specification, and the voltage gain calculated in dB using the following formula:

$$\text{Gain (dB)} = 20 \log (\text{Vout/Vin})$$

As a minimum, the gain shall be measured at 400, 1000, 1600, 2000, 2400, 3000 and 6000 Hz to insure conformance to the requirements of Requirement 13. The linearity requirement of Requirement 16 shall be checked at 1000 Hz by setting the amplifier input voltage to 50 microvolts and increasing it slowly to 500 microvolts. The amplifier gain shall remain constant within ±2 dB.

* c. Amplifier input power measurement. The power supply shall be adjusted so that 27.5 volts dc appears across the output terminals of the DC bias supply when placed in the test circuit. The frequency generator shall be set to 1000 Hz. The input current shall be checked with a DC current meter.

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d. Amplifier distortion measurement. Amplifier THD shall be checked under load conditions using one of the two methods specified for measuring transformer distortion in Requirement 10(c). The distortion shall be measured at several different frequencies in the range of 1000 to 3000 Hz or as otherwise specified to the satisfaction of the cognizant Government inspector.

* e Amplifier stability. The amplifier THD shall be checked for conformance with the requirements of Requirement 17. With the output terminated in the proper resistive load, the dc supply voltage shall be varied from 10 to 30 volts while measuring THD, and the load shall be removed and then restored to verify that no damage occurs. Amplifier stability need only be checked at 1000 Hz.

f. Amplifier input impedance measurement. With switch S20l still in the position so that R201 is out of the circuit, the signal generator shall be adjusted for a 1000 Hz, 500 microvolts signal at the input of the amplifier under test. Then switch S2Ol shall be set to include R_{201} in the circuit path, and R_{201} shall be adjusted so that the amplifier input voltage drops to 250 microvolts. The value of R201 shall be recorded as the amplifier input impedance.

20. Microphone Keying Switch. The microphone keying switch shall be a Micro-Switch (91929) type 2PB28l or equivalent (as approved by the procuring activity).

21. The stay cord shall be firmly attached to the switch assembly.

NOTES:

l. Features shown in dotted lines are for reference only. Item need not be constructed in this manner.

2. Dimensions in inches. Unless otherwise specified, tolerances:

Metal Parts + .005

Molded Sections ± .05

* 3. The parts specified on this specification sheet are for use on completed and tested cable assemblies Qualification of individual parts and components is not required.

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Navy - AS
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